

Lincoln College



ANNUAL
REVIEW

1967

UNIVERSITY OF CANTERBURY

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PREFACE

THIS *Annual Review* records events at Lincoln College during 1967. The most important event by far was the completion of the Hilgendorf Wing. This well-designed, utilitarian but aesthetically pleasant building will increase the efficiency of teaching and research, and by raising the morale of both staff and students will act as a magnet for talent at Lincoln. Mr J. H. Scott, Assistant Registrar, has written an article about the Hilgendorf Wing for this *Review*.

Two other important events occurred in 1967. First there was the proposal for a new degree. Professor R. H. M. Langer has written an article describing the changes which we hope will fit future graduates to a rapidly changing environment. The other important event was the arrival of a 1130 IBM computer at the College. Mr N. S. Mountier describes its place in the teaching and research of the College. Computers and mathematics will undoubtedly be a part of normal professional life in ten years' time and Lincoln graduates will be in a position to understand their use.

Dr I. D. Blair's article on the research work of the Microbiology Department indicates that behind the scenes and over many years, Lincoln has made a major contribution to this important field. This contribution has been in terms of research, and trained graduates who have made their mark.

A. T. G. McARTHUR

Senior Lecturer in Rural Education

EDITOR

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BUILDING DEVELOPMENTS AT LINCOLN



J. H. SCOTT

Assistant Registrar (Academic & Buildings)

THE eighth of March 1968 was a momentous day in the history of the College for, on that day, His Excellency Sir Arthur Porritt, Governor-General of New Zealand, officially opened the Hilgendorf Wing of the Teaching Block—the first building in a development scheme which is designed to provide facilities for the 2,000 students expected to enrol at the College by 1980. In the Hilgendorf Wing are, at present, housed all the departments of the College except the departments of Agricultural Engineering, Soil Science and Wool Science.

The first wing to be built was named the Hilgendorf Wing to commemorate the services to the College and to agriculture of the late Dr F. W. Hilgendorf who for thirty-six years was a member of the staff of the College, the last six as Professor of Agriculture, was Acting-Principal on three occasions and who, after his retirement in 1936, served for seven years as a member of the College Board of Governors.

The building named in his memory is a five-storey building with a gross area of 109,000 square feet. Designed for a student population equal to 550 full-time students, the building houses lecture rooms, undergraduate and research laboratories, staff offices and specialised facilities for the carrying out of research projects on a variety of topics in the departments housed in it.

The method of construction chosen for the building was flat slab reinforced concrete for the floors with post tensioning and concrete block for the internal partitions. So that the building will harmonize with existing College buildings, part of the walls and the wall panels are faced with brick while the upward curve of the roof

features links the building architecturally to the downward curve of the gables on Ivy Hall.

The aim of the architects was to provide for a given sum a fixed net area and every device to achieve a balanced economical solution was used. To keep costs to a minimum, maximum off-the-site fabrication and finishing was aimed for, together with repetitive units to gain the advantage of factory production. For example, the entire east and west walls are made up of identical window units each weighing $3\frac{1}{4}$ tons which were constructed off the site and brought to the College only when required for lifting into position. The use of precast window units for these two long walls allowed a rapid construction of the exterior. At the peak sixteen units were erected on a normal working day.

The student entry to the building is on the ground floor at the south end. Opening off a concourse and foyer are five of the eleven lecture rooms in the building. These five rooms have tiered seating and are heated by water heating coils embedded in the floor, supplemented in winter with warm filtered air.

On the first floor and again in the south end are the remaining six lecture rooms while the main student laboratories are on the floors above.

Access to the staff and research area is at first floor level and all occupied rooms are raised to this upper level to prevent nuisance at pedestrian level. The ground floor is then utilised for housing service rooms and cloak room and toilet facilities for students. The walls of the ground floor are set back to provide a covered access around the building. A pass through at ground floor level between the student entry and the rest of the ground floor prevents the building being a complete visual barrier on the site.

The services to the building are housed in ducts between window units and under each window sill. Access to the interior is through an opening in the window units and this permits easy change in room usage. The prefabricated concept is continued in the laboratories and all benching, under bench cupboards and wall units such as shelving, draining racks, and cabinets are easily removable and are interchangeable. The laboratories are designed on the "finger" principle throughout the building. The "fingers" are the laboratory benches springing from the service ducts within the external wall.

In the Plant Science Department are fourteen plant growth cabinets to provide the artificial climatic conditions necessary for satisfactory research in this department. On the roof are five conditioned glasshouses in which plant, microbiological and entomological material for classroom teaching can be housed and in which

research projects in Plant Science and Horticulture, Microbiology and Zoology, can be carried out.

Staff offices all have the same dimensions and it has thus been possible to ensure that all fittings are standardized and interchangeable.

Laboratories and offices are heated by hot water convector-type heaters. The water in the heating system is heated with steam carried in a service duct from the College boilerhouse.

Although the opening of this wing will give the College some relief from the acute accommodation problems with which it has been faced over the last few years this building, will not, on its own, solve the College's accommodation problems. Accommodation in the building was provided for the staff and students expected in 1972. The College's own estimate of the numbers of full-time equivalent students expected to enrol in 1972 was 630 but this figure was challenged and approval to proceed was given on the basis that no more than 550 full-time equivalent students would enrol by 1972. In 1963, however, it became apparent that even the College's higher projected enrolments were likely to be exceeded. The College, in its projections had made provision for increased rolls as the numbers of school leavers, the "post-war bulge", increased but the rolls increased even beyond this as the percentage of school leavers enrolling at the College increased. A special case was put to the Government for the size of the two biggest lecture rooms to be increased and this was approved but the building, apart from the increased size of two lecture rooms, is designed for only 550 full-time equivalent students. The College roll in 1967 was the equivalent of 620 full-time students and in 1968 it is 750! The College is therefore facing already the prospect of a building crammed to capacity in its first year of use and the prospect of further continued use of the so-called temporary buildings extremely likely. The problems of these rapidly increasing rolls can be seen when it is stated that even with the size of the largest lecture room in this wing increased as it has been it now seats only 180 students. The largest single class at the College in 1968 is 220. In 1960 amazement was expressed at the College's estimate of 260 for the total enrolment in Diploma classes. We now expect an enrolment of around 260 in one class in 1969.

The enrolment in full-time classes at the College in 1958 was 338 students. The roll in 1968 of 890 students represents an increase of over 160%. Over the same ten-year period the increase in rolls in the other universities is of the order of 100%. The increase in the five years from 1963 is around 40% in some of the other universities but 120% at Lincoln.

Although the increased rolls pose problems with the size of single

classes, particularly in the Diploma course, the increases are not confined to the Diploma classes. There has been a considerable increase of the numbers of students enrolling in degree classes and an encouraging number of those who complete first degrees at the College are enrolling subsequently in courses for Masters' and Doctors' degrees. In 1948 there were five students enrolled for post-graduate degrees, eleven in 1958 but in 1968 eighty-four.

The Hilgendorf Wing, large though it is, was not designed to cope with numbers of students of this magnitude and even before it was completed College staff have been preparing a case for a second wing parallel to this and linked to it in which can be provided teaching and research accommodation for those students and staff who are already so cramped in the Hilgendorf Wing.

Between the Hilgendorf Wing and the proposed new wing are to be two lecture rooms, one to accommodate 350 students and the other 300 students. Approval to prepare sketch plans for these two lecture rooms was given early in 1968 and it is hoped that they will be built ready for the 1970 session. When the new wing is built it, and the Hilgendorf Wing, will provide teaching and research accommodation for all the departments of the College except for the Department of Agricultural Engineering which is being developed alongside the New Zealand Agricultural Engineering Institute whose buildings are situated on the College campus.

The development of these teaching facilities and the essential ancillary facilities which must accompany the provision of teaching buildings is being planned on a central area of the College campus with Ivy Hall, in which the College's administrative facilities are housed, as the centre. The College Council has set aside 130 acres of this central area on which future development in the academic, research and associated fields will take place. As a first step in this development all farming and horticultural activities in this area are being moved from this 130 acres.

Teaching buildings, as has been described, are being concentrated on two areas while student, recreational and other amenities are being planned for an area round the present Refectory. The Conference Hall to be named the W. H. Gillespie Memorial Hall, in memory of the services to the College of the late Hon. W. H. Gillespie, M.P., is the first building planned for this area while a Sports Centre comprising gymnasias, squash courts and other sporting facilities is planned for siting on the eastern edge of the College sports field only a few yards from the Conference Hall and the Refectory.

Traditionally, Lincoln College has been a residential institution and up until 1963 all students who wanted to live in residence could

be allocated residential places. The rapid increase in enrolments over the past five years, accompanied as it has by only a slight increase in the number of beds available, has resulted in a rapidly decreasing percentage of students who can be accommodated until in 1968 there are only sufficient beds for about 40% of the student population.

In an attempt to increase the numbers of beds to a point where every student who wants to live in residence can do so the College has plans for two types of residential hall—a tower block having 84 students and a smaller two-storey “motel” type hall for senior students—which will be built in an area of the campus allocated for residential development. This area will be reasonably close to the Refectory and to the sports field and Sports Centre.

Even with the halls of residence proposed it will probably never be possible for the College to become wholly residential again and the College is therefore planning facilities for what is a relatively new phenomenon on the College scene—a large number of students who travel to the College daily. Most of the needs of these students can be met from within the recreational facilities to be provided but it is necessary for the College to provide some space in which these students can read or study in quiet between lectures. This is being done by utilising the space underneath the tiered seating of the proposed new lecture rooms as study space.

The opening of the Hilgendorf Wing marks the completion of the first stage of the College's plans to provide adequate facilities for the 2,000 students expected to enrol by 1980. What happens after the roll reaches 2,000 is not known as yet but the development plan for the College is so designed that expansion beyond 2,000 students as an agricultural institution solely or as a full university institution, if this is warranted, are both possible. The College is willing and able to play its real part in the educational future of New Zealand and is looking forward to the achievement of its aims with vigour, interest and enthusiasm.

A great deal of credit must go to the architects, Trengrove, Trengrove and Marshall, and to the contractors, M. L. Paynter and Co. Ltd, to whom the College is most grateful.

RESEARCH IN AGRICULTURAL MICROBIOLOGY



I. D. BLAIR

Reader in Microbiology

IN 1884, four years after the College opened, the first Director, W. E. Ivey, published a report on "The Prevention of Smut in Wheat" (N.Z. Country Journal 8, 179). This would likely be the first recommendation in this country on the value of seed disinfection, using at that time copper or formalin liquid steepes. Now at this present time 84 years later, the smut diseases are under good control and in recent years we at the College have made further research contributions in the development of seed disinfection practices. There are signs though that the smut diseases are recurring again in severity, possibly through that strange attitude that arises from time to time when proven control techniques are ignored or imperfectly used.

In 1895, the second Director of the College, J. Bayne, was the author of a report on "Experiments with Rust-Preventing Manure" (N.Z. Country Journal 19, 493). His "manure" evidently did not succeed and this disease which in the world picture is pestilential in wheat, has also in some New Zealand districts been increasing in prevalence. In the early 1930s J. W. Calder conducted field experiments on seed disinfection collaboratively with J. C. Neill then of the Plant Research Bureau and Calder was also responsible for guiding the first masterate students in plant pathology, viz. the author of this article and K. M. Harrow, now of Plant Diseases Division, Auckland.

In the twenty-year period 1936-55, ten masterate (M.Agr.Sc. and M.Sc.) students completed theses most of which were concerned with investigations in plant pathology. The work accomplished by

staff and students was predominantly linked with problems of disease in field crops as exemplified by the following:

The cereal root knot complex (*Fusarium*, *Ophiobolus* *Cercos-porella*): N.Z.J.Sci.Tech. 19, 1-21; 20, 156-63; 36A, 207-20; D.S.I.R. Information Bulletin 4. 94p.

Wheat foliage diseases (*Erysiphe* and *Septoria*): Ann.App.Biol. 37, 570-83; N.Z.J.Agr.Res. 2, 208-13.

The thesis of H. C. Smith on powdery mildew led the way to the achievement of himself and co-workers in the Crop Research Division, D.S.I.R., in producing by breeding methods, mildew resistant varieties of wheat. (N.Z. Wheat Review 8: 32.) This resistance however has not persisted and severe mildew disease in 1967-8 was rampant in many parts of Canterbury.

Investigations were also brought to the point of publication in the following instances:

Disease of oats: N.Z.J.Sci.Tech. 37A, 526-34.

Pea diseases: N.Z.J.Sci.Tech. 31A, 17-23.

Problems of seed disinfection: Proc. 2nd Int. Congress Crop Protection, London 1949; N.Z.J.Sci.Tech. 32A 1-21; 34A, 397-404.

Loss and damage in potatoes: Tech. Publication 18, Lincoln College 24p; Proc.N.Z.Inst.Agr.Sci. 6: 104.

Associated with the above and later plant pathological work were K. M. Harrow, D. A. Shakleton and S. A. Menzies now with the Plant Diseases Division, D.S.I.R.; H. C. Smith now Director, Crop Research Division, D.S.I.R., K. B. Ram and A. R. Patwari now with research institutes in India; E. C. Wratt, farming in Marlborough; I. A. M. Cruickshank, with C.S.I.R.O. Canberra ACT and H. T. Wenham, Massey University.

The research on crop diseases has been of substantial order but our concepts on the wheat problem did not go beyond the pathology and ecology of root rot fungi. Although it was clear that the action of these did not fully explain yield loss, we did not make the breakthrough. This was brilliantly expounded in 1955 by H. C. Smith, then working in the Plant Diseases Division, D.S.I.R. when he showed (Commonwealth Phytopath News 1: 10-11) that the aphid transmitted Barley Yellow Dwarf Virus was the major problem in our wheat growing. His numerous later papers have shown the effectiveness of spray treatment in control and this technique has proved as strikingly effective in aphid control as seed disinfection in smut control. These points are made at a little length for the reason that the wheat disease studies at Lincoln reveal an evolutionary pattern of development.

At the present time the scope of plant pathology in the department has altered. Occasionally we have reported on a topic of a fundamental botanical nature e.g. Trans.Roy.Sco.N.Z. 88, 231, but the dominant activity is engaged with a study of deterioration in lucerne supported by a D.S.I.R. research contract. The activity of various disease organisms has been surveyed through all lucerne districts in the South Island. There is evidence however, of the significance of non pathological factors limiting lucerne thrift, viz. nutrition, and this work appears likely to develop into an investigation of a general agronomic nature.

The significance of disease in limiting the development of Sweet Brier (*Rosa* sp) is also being studied. *Fusarium* fungi have on reinoculation into healthy plants been shown to produce marked restriction of brier growth, possibly associated with metabolic products of fungus growth but this factor has not been defined.

With the appointment of Dr R. F. R. McNabb, whose work on Basidiomycete fungi is widely known, it is expected that an impetus will be provided to the plant pathological work.

The next phase of development was in accord with the interest in the early 1950s of some plant pathologists in other parts of the world in developing a closer emphasis of research on an understanding of disease phenomena linked with the ecology of soil organisms. Our first contributions in this area of research were: *Annals.App.Biology*, 30, 118-27; *Canadian J.Research* C20 174-85.

The trend of interest however, was away from plant disease organisms. R. H. Thornton, now with Soil Bureau, D.S.I.R., in a masterate thesis reported on the occurrence of soil actinomycetes and subsequently from his postgraduate experience in Britain and work in the D.S.I.R., became known as an authority on the activity of soil fungi and one of the discoverers of the fungus cause of the facial eczema disease. (*N.Z.J.Agr.Res.* 1, 922-38; *Nature* 182, 1095-6.)

Miss M. N. Crozier surveyed the occurrence of nitrogen-fixing bacteria in clover but as is often the case in a research study was able to observe an unexpected feature of fundamental interest, viz. the occurrence of bacteria, other than *Rhizobium* nitrogen-fixers, in healthy clover tissue. (*Canadian J. Microbiology* 3: 125.)

There was a marked stimulus arising from the two-year stay here on a Commonwealth Scholarship of J. B. Robinson (1960-2) from Ontario Agricultural College, Canada. His programme was comprehensive and led to an elucidation of several, hitherto unreported aspects of the microbiology of soils under tussock grassland. (*Plant and Soil* 19, 173-83; *N.Z.J.Agr.Res.* 7, 569-82.) Interesting and intellectually valuable controversy arose from some facets of Robinson's published research when he, using different methods from



AT LINCOLN COLLEGE

Miss Dhasna Subhanij, a degree VI student from Thailand under the Colombo Plan, talking to Mr Bill McKeown. The new Hilgendorf Wing is in the background.



THE NEW HILGENDORF WING FROM THE COLLEGE GATES

The 110,000 sq. ft. building was designed by Trengrove, Trengrove and Marshall and built by M. L. Paynter and Company Ltd.



THE COLLEGE REFECTORY

The cactus garden on the wall of the refectory.



The College caters for over 800 students for lunch.



THE IBM 1130 COMPUTER USED FOR RESEARCH AND TEACHING

Miss E. Emerson operating the computer. The control processing unit is in the foreground, and in the background is the line printer.

other workers (N.Z.J.Agr.Res. 8, 723) questioned the veracity of some data on soil biology earlier published in New Zealand scientific literature. The controversy was a stimulating one and features of it have afforded admirable material for presentation to research students, both on technical and philosophical criteria.

From Robinson's influence and impact, others were drawn to enquire into the microbiology of soils of a primitive nature, undisturbed by agricultural activity. A thesis by A. H. Nordmeyer, now farming, and by P. E. Hollings, now a postgraduate student in Guelph, Canada, revealed the interest of this kind of study in the soils of the upper basins in the Craigieburn mountain range and on the chronosequence sites below the face of the Franz Josef glacier.

The earlier interest in *Rhizobium* nitrogen-fixing bacteria has been maintained and aided by research assistance from Miss A. Bennett, Miss R. Shroff, and B. N. Clayton, the significance of strain in *Rhizobium trifolii* and *Rhizobium meliloti*, was demonstrated. Furthermore, some agronomic aspects of inoculation and nodulation were demonstrated. (N.Z.J.Agr.Res. 3, 804-19; 10, 66-81; 10, 237-52.) At this stage in this work, Mrs V. Price from Ohio, U.S.A. is collecting *Rhizobium* bacteria from diverse ecological situations where selfsown medics and clovers have become established, particularly in unimproved tussock grassland. These bacteria are being assessed and the strain characteristics defined, as part of a search for *Rhizobium* inoculant organisms for use on problem soils where there is a need to surface sow clover and lucerne.

Soil microbiology has also now been expanded in concept by the projects for which Dr A. P. Mulcock has attracted financial support (N.Z. Forest Service, and Ivon Watkins-Dow Ltd).

The Ph.D. programme of M. Noonan is based on the circumstances in some Nelson pine forest soils where the growth of the first *Pinus radiata* crop being satisfactory, contrasts with failure of tree growth in some second rotation stands. Within the nutritional complex likely involved, two forms of microbiological study are in progress. First, a survey of organisms associated with decomposition of pine residues and second, studies on mineralisation of carbon and nitrogen.

With Dr Mulcock, S. H. Manning is studying the interaction of soil organisms with toxic herbicide substances such as picloram (Tordon). Trial sites of treated and untreated soil are being sampled for a microbiological analysis, while the physiological activity of organisms under the impact of Tordon treatment is being measured by perfusion and microrespirometry.

Dr Mulcock and Miss C. A. Reay have also made progress on a project concerned with demonstrating the utilisation of sawdust for cultivating edible mushroom fungi.

The initiative of Dr Mulcock in opening the subject of microbiology of sheep fleece wool under New Zealand conditions has led to some particularly interesting developments. With I. E. B. Fraser, now of the Wool Research Organisation of New Zealand Organisation of New Zealand a report was published in 1956 on the occurrence of pigment stained wool arising from bacterial action. (Nature 177, 628-9.) This report was followed by others on the nature and significance of bacteria in the fleeces of typical flock sheep. (Australian J.Agr.Res. 9, 704-7; N.Z.J.Agr.Res. 9, 1 16, 485-96.) Dr Mulcock's research group have continued to make progress on this topic and with Mrs P. E. Horn a report on thermophilic bacteria in wool was recently made. (N.Z.J.Agr.Res. 8, 818.)

In collaboration with Mrs M. Hoare, a clear picture has now been obtained of the extent and nature of pyocyanine production in wool by *Pseudomonas aeruginosa*. Microbiological causes of brown, mauve and yellow stains are also being studied. Earlier Dr Mulcock had demonstrated degradation of wool fibres through the endophytic growth within fibres of a fungus *Peyronella glomerata*. (Australian J.Agr.Res. 16, 691-7; N.Z.Vet.J. 13, 87-93.) This kind of physical defect is still attracting interest with attention now to the activity in fibre breakdown by bacteria like *Bacillus subtilis* and D. N. Brewer, Veterinary Department, is a collaborator in this work.

Research contributions have also been made by Dr Mulcock in the study of microbiological contamination of meat and his work has provided some material for his participation in the courses of instruction for the College Certificate in Meat Inspection. (N.Z.J.Agr.Res. 8, 165.)

Dr Mulcock and Miss B. Peddie are currently engaged on a comprehensive investigation relating to the microbiology of some poultry foods. The facts of physiological disorder and disease in young poultry fed on cereal mashes has attracted notice and concern and the New Zealand Poultry Board has provided financial support for this research work being done in the department. The objectives include identification of fungi and bacteria isolated from grain mash and the examination of the role of several of them in contributing to forms of toxicity detrimental to the health of birds. Feeding trials are in progress and histological investigations on changes occurring in the tissues of affected birds are being undertaken.

It is for others to appraise the merit of the research outlined herein but at least it can be said agricultural microbiology at Lincoln has been of interesting and dynamic scope. The varied projects, most of which have proceeded to the level of publication in research

journals have also afforded training to many people who from the start at Lincoln have become successful research workers in other places. Our lecturing programmes are also adequately illustrated in reasonable part by the evidence derived from our own investigations.

NEW DEGREES AT LINCOLN



R. H. M. LANGER

Professor of Plant Science

ANY university which is worthy of this name must be sensitive to change, both within its own four walls and outside in the community it serves. It must be dynamic and alive. It must be prepared to respond to changed circumstances and, if possible, be ready to meet them before they arise. Applying this dictum to Lincoln College we are conscious that the last ten years have seen a very considerable transformation of the local and the national scene. At the College itself, we have witnessed tremendous growth together with greatly increased emphasis on the many specialised disciplines which jointly make up the composite subject called Agriculture. One need only compare the 1967 staff list with that of ten years ago, or contrast the Hilgendorf Wing with the motley collection of old temporary and semi-temporary College buildings, to appreciate the magnitude of this change. No matter how soundly our reputation was founded in the past, we can say that Lincoln College has now grown into a teaching and research organisation of international calibre. This almost explosive evolution has been engendered not only by ever-rising numbers of students, but also by an increase in the multiplicity of demands made upon us. As time goes on, we find more and more outlets for our graduates, in addition to the main needs which we continue to service. Lincoln graduates are wanted for a great variety of fields and occupations, and naturally we have to be aware of these requirements and meet them to the best of our ability. From all this it follows that in order to cater for changing conditions within and outside the College we have to revise our course structure periodically.

One of these major revisions has just taken place, and all new students entering the College in 1968 will be completing their

courses under new regulations. The old degree will be phased out, in the sense that all students who enrolled in 1967 or earlier will follow the previous course structure until they have graduated.

The major objective in revising the degrees was to retain our strength in the field of general agriculture but, in addition, to cater more adequately for specialised interests in agricultural science and business. In practical terms this meant that we have kept the present Bachelor and Master degrees in their broad essentials but for the more able and successful students with specialist leanings we have provided a new avenue in the shape of a B.Agr.Sc. and a B.Hort.Sc. with Honours. This Honours course occurs in the fourth year of the degrees but it is foreshadowed by the grouping of students half way through the third year into a business, a science and a general stream. A student who obtains first or second class Honours at the end of four years is eligible to proceed directly to a Ph.D. At present our students have to take the M.Agr.Sc. with Honours before they may enroll for a Ph.D. This often means another one and a half to two years of work (although it need not take as long) and also greater difficulty in competing for post-graduate scholarships because our students are normally not ready to apply at the end of the year when the University Grants Committee makes its awards. Very few agriculture students figure in the award lists to the detriment of the country as a whole. Apart from this major innovation, the opportunity has been taken to revise prescriptions, and in particular an attempt has been made to broaden the scope of the introductory courses by including aspects of agricultural significance. Thus, the physics of the earth and the solar system or the botany of agricultural plants will be taught in the first year. This replaces the classical courses in Physics and Botany which we inherited from pure science degrees. Several subjects have been renamed. Statistics appear as a full subject in the second year and there are several other important changes.

The first year of the B.Agr.Sc. contains four subjects—Biological Science Ia and Ib, and Physical Science Ia and Ib, all of which are the modified counterparts of our present Degree I subjects. In the second year we have Biological Science IIa which contains Biochemistry and Animal Physiology, and Biological Science IIb made up of Microbiology, Entomology together with a second paper dealing with Genetics, Ecology and an introduction to crop production. The other three subjects in the same year are Soil Science I, Agricultural Economics I and Statistics.

The third year of the B.Agr.Sc. will see more significant changes. In the first half of the year we will continue to have Animal and Plant Science, under the new name of Agriculture A, together

with Farm Management Ia. However, at the half-year mark students will be asked to make the decision of whether they wish to continue with a fairly general course (Group A), or alternatively whether they wish to give more emphasis either to the business aspects of agriculture (Group B) or to agricultural science (Group C). Irrespective of which group they choose, all students will, in the second half of the year, take Agriculture B which contains selected aspects of Animal and Plant Science, Animal Health, Soil Science and Agricultural Engineering. Students in Group A will also take Farm Management Ib, Agricultural Economics IIa and Agricultural Engineering, so that their course is not greatly altered from the present pattern. On the other hand, Group B students will replace Agricultural Engineering by a new subject called Agricultural Economics IIb which deals with price and demand theory and with production theory. Group C students combine Agriculture B with a composite subject, called Biological Science III which covers modern biological techniques both in theory and in practice. Pest control is included for all students in Agriculture A and disease control in Agriculture B.

According to the new regulations, any student who has reached a sufficiently high standard in the first three years may be permitted by the Professorial Board to enrol for an Honours course in the fourth year. Selection of such students thus lies in the hands of the Professorial Board. If selected for Honours, the student will be allocated to the department in which he wants to study. He will then offer one subject in which he will sit three examination papers together with a general essay paper covering the broad principles of his field. Honours students must also undertake a project which has to be submitted by 31 October of the year in which he takes his examinations. In other words, the M.Agr.Sc. thesis has been replaced by a much more modest and shorter exercise, to be completed in the same year. If successful, these students will be awarded the B.Agr.Sc. with first or second class Honours after a total of four years of study. This may encourage them to proceed to a Ph.D. and to take their rightful place among the applicants for post-graduate awards. These normally entail taking the Ph.D. in New Zealand followed by a postdoctoral year overseas.

Those students whose record after three years is not good enough for them to be permitted to go for Honours, will take four subjects in the fourth year. These subjects have to be taken from the group to which the student belongs but, as at present, the General Agriculture Option (Group A) contains a great deal of choice. Each student will be supervised by the department which covers his major interest. At the end of the fourth year successful students will

graduate B.Agr.Sc. without Honours in exactly the same way as they do now. If they want to continue their studies they may enrol for a Master's degree which will be awarded with or without Honours just as at present. In other words, this group of students follows a course which differs in no essential feature from the present degree structure. A further point is that the student who matures slowly can, if he is able enough, achieve the same position in the end as his more advanced colleagues.

The only other new feature of the new degree structure is to draw attention to exemptions from subjects, normally in the first year, in the case of students who come from other universities or who have done well in the Scholarships Examinations. This may help to attract bright students to come and study at Lincoln College. The degrees in Horticulture are patterned on the B.Agr.Sc. in that provision has been made for a B.Hort.Sc. with Honours along similar lines with appropriate replacement of subjects. Thus in the second year instead of Animal Physiology aspects of horticultural botany including the physiology of cultivated plants are being taught. Half-way through the third year students are grouped in three streams as for the B.Agr.Sc. leading to more intensive studies in horticultural business, science or general horticulture. Within these streams further selection is possible in the fourth year and for selected candidates an Honours course in one subject may be approved.

The main innovations of the new degrees are that an attempt has been made to create a structure which follows more closely than before the modern trends in other Professional Schools of the University system. Without sacrificing our important role as the training ground for the broadly based advisory officer and farm manager, we hope that we have catered more adequately for an increased output of agricultural technologists both in science and in the business sphere on whose contribution the future progress of the agricultural and horticultural industry so heavily depends.

It has taken over three years of discussion and negotiation to create the new degree structure. Lincoln College is a very complex organisation and consequently it has been difficult to cater for all legitimate requirements. Some features of the new degrees may fall short of what was hoped for. But, on the whole, I firmly believe that we have taken a bold and important step forward into the future. Let us hope that in years to come we will look back on the introduction of this new scheme as a far-sighted and imaginative decision in the evolution of Lincoln College.

THE LINCOLN COMPUTER



N. S. MOUNTIER

Senior Lecturer in Biometrics

ONE of the most far-reaching developments in technology has been the electronic computer. Its possible uses are extraordinarily great, and society has yet barely begun to feel the repercussions of this strange and versatile tool.

Basically, what a computer does is simple enough. It performs arithmetic operations at enormous speed and can do hundreds of thousands of additions per second. As well, it can be pre-set or programmed to carry out an extensive series of operations, choosing from alternative paths according to the results it has already calculated. It is the applications of this that are so great: there are the more obvious applications in science, engineering and business in which extensive calculations are required. But as well as this, the computer is being used in such diverse fields as language translation, medical diagnosis, testing authorship of books (notably the New Testament epistles), the control of industrial processes (known as automation), the control of space vehicles, and seat-booking for airlines.

Computer manufacture is still advancing at an enormous rate and applications are still widening. Over the next forty years everyone will be influenced by computer usage, and in particular, every agricultural scientist will make extensive use of computer applications. The students we are educating now are going to be working in the next forty years, and it would be a grave omission to neglect to give them some acquaintance with the uses and limitations of the electronic computer.

A computer is very fast and very accurate, but it has no initiative whatever. It must be instructed to do every step that a calculation requires, and the *instructions* must cover every possibility that can

possibly arise. The computer does what it is told to do, simply and completely. A child who does literally what you say, rather than what you mean, can be infuriating, and a computer always does exactly what you say. A whole set of instructions is called a *program*, and it is only when a suitable program is written that a job can be done on the computer. The effective controller of the computer then is not the person standing over it pressing the buttons, but the person who wrote the program that it is operating.

The Lincoln College computer (an IBM 1130) was installed at the beginning of September 1967. It was put in the new Hilgen-dorf Wing, and was in fact the first occupant as the building was not completed until November. The fields in which the Lincoln computer is being used are education, research and administration. The major emphasis so far has been on research, but uses in the other areas are developing.

This year, a course on computer usage is being given to all fourth-year degree students and to students doing Dip.Agr.Sc. This is a short course (less than twenty hours in total) and its aim is to make students acquainted with the computer, so that they understand something of how it works and how they can make use of it. The course includes some instruction in programming because this is one of the ways of understanding the use and the limitations of a computer. (It could also be argued that computer programming is a valuable exercise in training logical thought.) At the end of this course, students will not be trained programmers, but they will have a familiarity with computers and a basis of knowledge on which other courses may build.

Teaching students about computers is not the only educational use of a computer. It can also be used as a teaching tool in other fields. For instance, in some exercises involving numerical work a teacher may be concerned that only a few students actually do the work, and the others simply copy their answers from the few. This can certainly be eliminated by making the figures different for each student, but would make the marking an impossible job by older methods. With a computer, however, each student can be set the problem with different data, and the computer can easily calculate the result of each set. Indeed, when the marked exercise is returned, a computer-printed model answer can be given to each student.

On a higher plane, some aspects of genetics and ecology depend upon random effects. A population of insects may flourish or decline partly according to chance. The computer gives a method of simulating the progress of populations over and over again, so that we can see in what proportion of cases they will flourish. This

technique of simulation is one made possible by the computer. It can be used only because of the computer's extraordinary speed of calculation. It is basically a research technique, but it is easy to see that it can also be used as a teaching technique where simulations will illustrate arguments in a very convincing way. To paraphrase an adage, "one simulation is worth 10,000 words"!

Computers are used in business even more than they are in science, and the administration of Lincoln College is quite a large-scale business. The Lincoln computer was acquired primarily for teaching and research, but there are a number of the Registry's tasks that will be done by computer in the future. The first to be dealt with is the analysis of student records. A start has already been made on this and this year's class rolls were prepared by the computer. The records of students' school qualifications and of their progress through university are the basic data on which various educational research must be done. If the value of the university entrance qualification is questioned, then records of the entry qualifications and relative progress of students are an obvious place to look for an answer. These records exist in written form, but it is difficult to extract the relevant data from piles of hand-written forms. Once these data are put on punchcards and fed into the computer all sorts of tables of comparisons can readily be printed.

Education and administration are both important fields for computer work, but the real impetus for the acquiring of the computer came from the research field. Some members of Lincoln College staff have been using an electronic computer in their research projects for a number of years.

This work was done at first on the University of Canterbury computer at Ilam, and, more recently, on a commercial computer in Christchurch. The pioneer computer users at Lincoln have been the Economics and Farm Management groups.

Economists are much concerned with predictions; predictions of production, predictions of prices and markets, and this work generally means dealing with very large amounts of data. You may think of hundreds of different factors that *might* affect the price of New Zealand lamb in Britain. The economist must try to sort out which of these factors are the relevant ones, and then use these to make his prediction. This process involves very lengthy calculations which might take months to carry out by hand. The computer will do such calculations in a matter of hours.

Economists and others are also given to carrying out surveys in which information is collected, usually on a fairly large scale, to find out the state of affairs in fertiliser usage, or wheat growing or lambing losses or vegetable prices or any of a thousand topics. Such

surveys produce a large mass of data, and the adequate digestion of these data is often impossible without a computer.

Farm management specialists do much work with a technique known as "linear programming" (a phrase which is quite distinct from computer programming). They are concerned to calculate the way in which a farmer may most profitably use the resources at his disposal—should he crop more or run more stock, how much capital should go into fertiliser, how much into stock and so on. The answers to these questions again require very extensive calculation and a computer is essential for most of this work.

Engineers also deal with masses of data and extensive calculations, and the hydrological field yields stream flows, rainfall and run-off data in great quantity.

On the biological side, computer work ranges through analysis of experiments, the computation of quality indices for sheep, insect population studies, bioassay, phosphorus uptake studies—to mention those more or less under way at present. As time goes by, work will be done on some of the more complex techniques of multivariate analysis in plant ecology. Taxonomy is another field on which the computer can have a marked impact, and there are many other topics in which the computer can be used as a very powerful tool.

CONFIGURATION AND ORGANISATION

The Lincoln College computer has a core store of 8,192 16-bit words, and a backing store of one magnetic disk which takes 512,000 words. Input is in a card-reader-punch which reads 300 cards per minute. The main output is on a line printer which has a speed of 80 lines per minute. Alternative modes of output are the console typewriter and the card punch. Subsidiary equipment consists of a card punch, a verifier and a sorter.

The Computer Laboratory is not a part of any College department, but is separately established with a Director (who is part-time in that he is also Senior Lecturer in Biometrics) and a full-time graduate Computer Programmer. Policy is guided by a special sub-committee of the Professorial Board.

RESEARCH IN PROGRESS



AGRICULTURAL ECONOMICS

(AGRICULTURAL ECONOMICS RESEARCH UNIT)

1. MARKETING ECONOMICS

Price and Income Elasticities of Demand for Meat and Dairy Products in Great Britain, B. P. Philpott, M. J. Matheson.

Estimate of regional elasticities have been derived and the implications of regional differences for promotion policy are being studied.

Long-Term Projections of Overseas Demand for New Zealand Meat, B. P. Philpott, M. J. Matheson, D. R. Edwards.

Projections are being prepared for Great Britain, E.E.C. countries, U.S.A. and Japan.

Price and Income Elasticities of Demand for New Zealand Wool, B. P. Philpott, M. J. Matheson.

An econometric model is being developed to explain short-run speculative fluctuations in prices and a simulation study of marking has been commenced.

Long-Term Projections of Demand for New Zealand Wool, B. P. Philpott, M. J. Matheson.

Trends in retail consumption and exports of wool textiles are being analysed for each of twenty-one major consuming countries. Individual country analyses will be used to project demand levels in 1970.

Economics of Private Wool Selling in New Zealand, L. D. Woods.

An evaluation of the role of private wool dealers, the advantages to farmers of selling wool by this method and the returns achieved. A survey of all dealers is being conducted.

Demand for Pigmeat in New Zealand, C. A. Yandle.

Time series analysis and a questionnaire survey are being used to establish factors affecting consumption of pigmeat. Long term projections of demand will be formulated. A critical analysis of marking methods is also being conducted.

Economics of Retailing fresh Vegetables with Special Reference to Supermarkets, R. W. M. Johnson, B. P. Philpott, and E. W. Kitson.

Surveys are being conducted in metropolitan areas to analyse differences in marketing costs in prices and pricing policies in major retail outlets for fresh vegetables.

2. PRODUCTION ECONOMICS

Economics of Hill Country Development, R. W. M. Johnson, D. D. Hussey.

This project is designed to measure the profitability of hill-country development and to assess the economic impediments which prevent farmers from

embarking on development programmes and to suggest measures needed to overcome these impediments such as tax policies and financing arrangements. The profitability of development on a group of Meat and Wool Board Economic survey farms has been studied and a survey of Taranaki hill country farms is being made. Special attention is being paid to the impact of falling prices on development.

Economics of Large Scale Land Development, R. W. M. Johnson, D. D. Hussey, H. J. Plunkett.

Economic analysis of development on Lands and Survey blocks with special reference to the capital requirements for development.

Economics of Beef Production in Hawkes Bay, R. C. Jensen, D. McClatchy.

Linear programming studies of profitability of beef on some North Island properties. Supply functions for beef. Comparative economics of beef from run cattle compared to dairy beef.

Supply Projections for New Zealand Agriculture Products, R. W. M. Johnson, D. McClatchy, B. J. Ross.

A recursive programming model for supply projection of wool is being built up and will be generalised for other farm products.

Economics of Farm Labour, J. L. Morris, R. G. Cant, A. T. G. McArthur.

A survey of farm labour problems in Cheviot County has been completed and is being analysed.

Economics of Fertiliser Use in Southland, R. C. Jensen, A. C. Lewis.

Economic analysis of case studies and survey farms.

Economic Implications of Agricultural Development in Northland, A. T. G. McArthur, B. P. Philpott.

The aim of this project is to assess the agricultural potential of Northland: to calculate the profitability of development and the overall economic implications for the area.

Econometric Analysis of Land Sale Prices, R. W. M. Johnson, D. D. Hussey.

Development of models using data on land sales.

Long-Term Trends in Agricultural Productivity, B. P. Philpott.

Trends in capital and labour use and the growth of efficiency in agriculture over past decades are being studied in order to elucidate the relationships between resource use and output achieved in the past. Time series production functions have been fitted to the data and results are now being prepared for publication.

Cross-Section Production Functions for a Sample of Sheep Farms, B. P. Philpott.

Combined cross-section and time-series production functions are being fitted to a sample of 300 sheep farms from the N.Z. Meat and Wool Boards' Economic Service.

3. INTERINDUSTRY ECONOMICS

Development of an Eight-Sector Interindustry Table for 1964/5, B. P. Philpott, B. J. Ross, J. W. Francis.

Using the 1954/5 and 1959/60 Statistics Department Input-Output tables, an estimate of the structure of the economy in 1964/5 and the trends in gross and net output, primary inputs, and final demand, is being prepared.

A Fifteen Sector Model of the Agricultural Economy, B. P. Philpott, B. J. Ross.

The agricultural sector in the eight sector table is being disaggregated into a fifteen sub-sector table distinguishing between different types of agriculture.

Capital Employment and Output by Sectors 1955/1965, B. P. Philpott, J. W. Francis.

An analysis of the relationship between output, capital labour and technical change in each of eight sectors and for primary produce processing industries. Parameter values will be used for projecting future capital requirements. Estimates of capital employed in manufacturing has been completed and work is proceeding on other sectors.

Projection of Structure of N.Z. Economy to 1974/5 under conditions of four per cent Agricultural Growth, B. P. Philpott, B. J. Ross.

Projections based on a pilot, four sector, model have been calculated and will be used as the basis for work in the larger model.

Indirect Employment Generated by Agricultural Production, B. P. Philpott, B. J. Ross.

A pilot assessment has been carried out using a four sector model and the methods used will be incorporated in the work involving a larger model.

PUBLICATIONS

37. Court, R. H., "Some Projections of Retail Consumption in New Zealand". Lincoln Coll. Agric. Econ. Res.

38. Morris, J. L. and Cant, R. G., "The Nature and Extent of the Farm Shortage in Cheviot County, Canterbury".

39. Frengley, G. A. G., "Index to N.Z. Agricultural Publications 1964 extracted from Index to N.Z. Periodicals".

40. Johnson, R. W. M., "High-Country Development on Molesworth".

41. Philpott, B. P. and Ross, B. J., "Input-Output Models for Projecting and Planning the Economy".

42. Philpott, B. P., Acland, H. T. D. and Tairo, A. J., "Statistics of Production, Trade Flows and Consumption of Wool and Wool-type Textiles".

43. Yandle, C. A., "A Survey of Christchurch Consumer Attitudes to Meat".

44. Jensen, R. C. and Lewis, A. C., "Fertiliser and Production on a Sample of Intensive Sheep Farms in Southland 1953-64".

45. Sanderson, K. T. and McArthur, A. T. G., "Computer Methods for Development Budgets".

46. Jensen, R. C. and Lewis, A. C., "Budgeting Further Development on Intensive Sheep Farms in Southland".

47. Johnson, R. W. M., "The Impact of Falling Prices on Taranaki Hill-Country Development".

Court, R. H., "Supply Response of New Zealand Sheep-Farmers". Economic Record, June 1967.

Jensen, R. C., "Economics of Pasture Insect Control". Proc. 20th N.Z. Weed and Pest Control Conf. 1967.

Johnson, R. W. M., "Income and Productivity in Agriculture". Proc. Linc. Coll. Farmers' Conf. 1967.

Morris, J. L. and Cant, R. G., "The Labour Position on the Farm". Ibid.

Ross, B. J., "Income and Output in New Zealand Agriculture". Canty. Ch. of Comm. Agr. Bull., August 1967.

Johnson, R. W. M., "Recent Price Trends in Farming". Ibid. November 1967.

AGRICULTURAL MICROBIOLOGY

PLANT PATHOLOGY

Deterioration of Lucerne, I. D. Blair, P. E. Burnett, J. Field-Dodgson

The work in this research contract with the Department of Scientific and Industrial Research has entailed field surveys of all South Island districts with the recording of data on plant population and recognisable diseases. Interest is mostly directed to *Verticillium* wilt, *Fusarium* and *Stagonospora* root rot, bacterial stem blight (*Pseudomonas medicaginis*) *Ascochyta*, *Colletotrichum* black stem, *Pseudopeziza* and *Stemphylium* leaf spot. Among 450 crops examined no evidence of bacterial wilt (*Corynebacterium insidiosum*) has been observed. There is evidence of widespread deterioration of the lucerne crop through physiological rather than parasitic causes, notably forms of nutrient deficiency.

Disease in Sweet Briar

Decline and death of sweet briar (*Rosa* sp) has been observed in some tussock grassland areas over a number of years. *Fusarium* fungi isolated from root tissue, have on reinoculation into glasshouse raised healthy plants, revealed a decline comparable with the field condition.

MICROBIOLOGY OF WOOL

Work over the last decade at Lincoln College has shown that the fleece of sheep has a microbial population as distinct as that of soil, milk or meat. The organisms present in fleece wool change both in numbers and species with change in the condition of the fleece. Moisture from rain and food from skin exudations are two examples of important factors but seasonal fluctuations also cause considerable changes.

The micro-organisms in the fleece do not cause any untoward effects under normal conditions but should certain species build up in numbers discolouration to the fibres may be produced or attacks on the fibre structure may cause sufficient weakening for breaks to occur in the fibres.

Topics at present being studied include:

Pigment Production by *Pseudomonas aeruginosa*, the cause of green stain of fleece wool.

Fluctuation in numbers of micro-organisms in the fleece of animals from two research flocks of Corriedale sheep.

Isolation and purification of a mauve pigment found in the fleece.

The identification and characterization of the cause of brown stain of fleece wool, A. P. Mulcock and Mrs M. M. Hoare.

Examination of the cause and factors affecting the incidence of shedding of wool and epidermal infections, A. P. Mulcock and D. N. Brewer.

STUDIES IN BIODEGRADATION

Micro-organisms are implicated in the breakdown of very many compounds both natural and synthetic. The mechanics involved and the factors which affect these transformations are of great interest. A number of topics are at present being studied.

The Breakdown of Organic Matter in Forest Soils, A. P. Mulcock and M. Noonan

Factors Affecting the Degradation of Pecloram ("Tordon") in Soil, A. P. Mulcock and S. M. Manning.

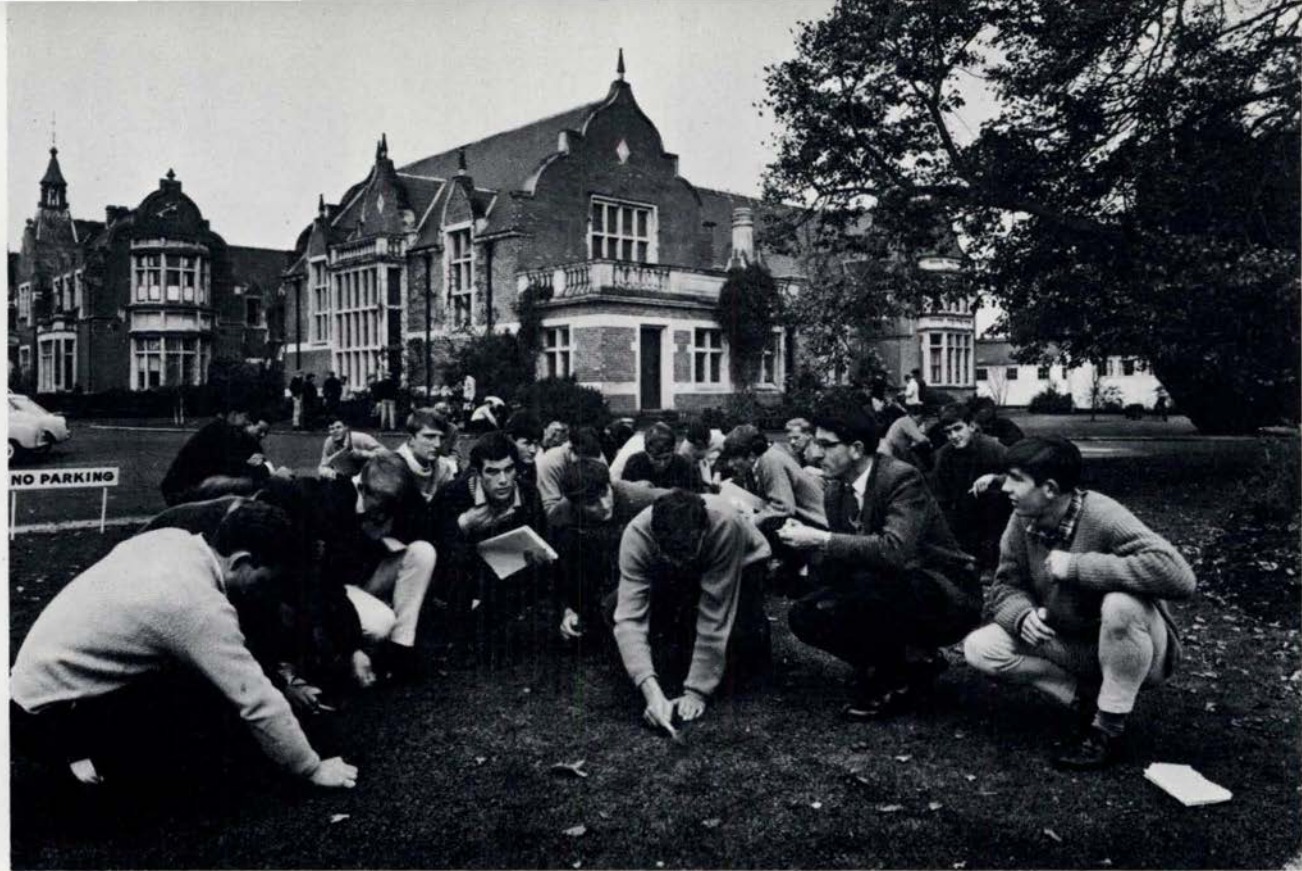


PLANT PHYSIOLOGY RESEARCH

Mr Brian Smith and Mrs Margaret McPherson working on research in the Old Laboratory Building.

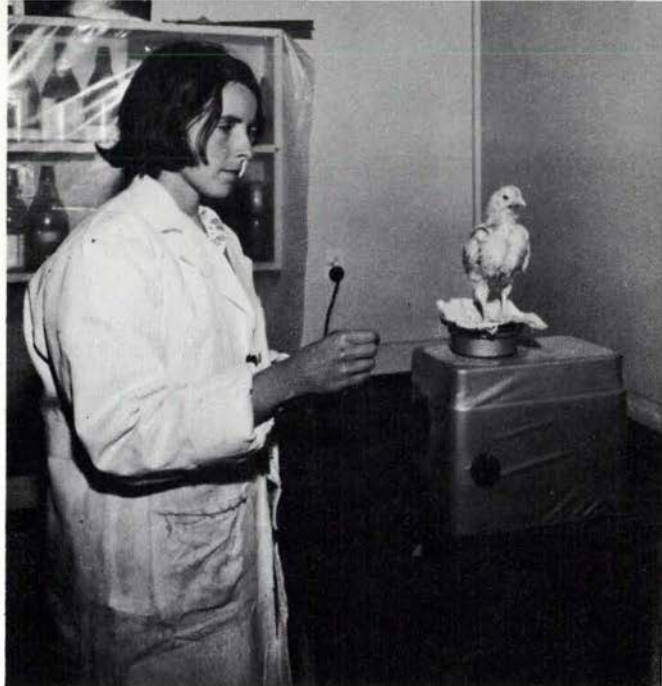
WOOL SCIENCE RESEARCH





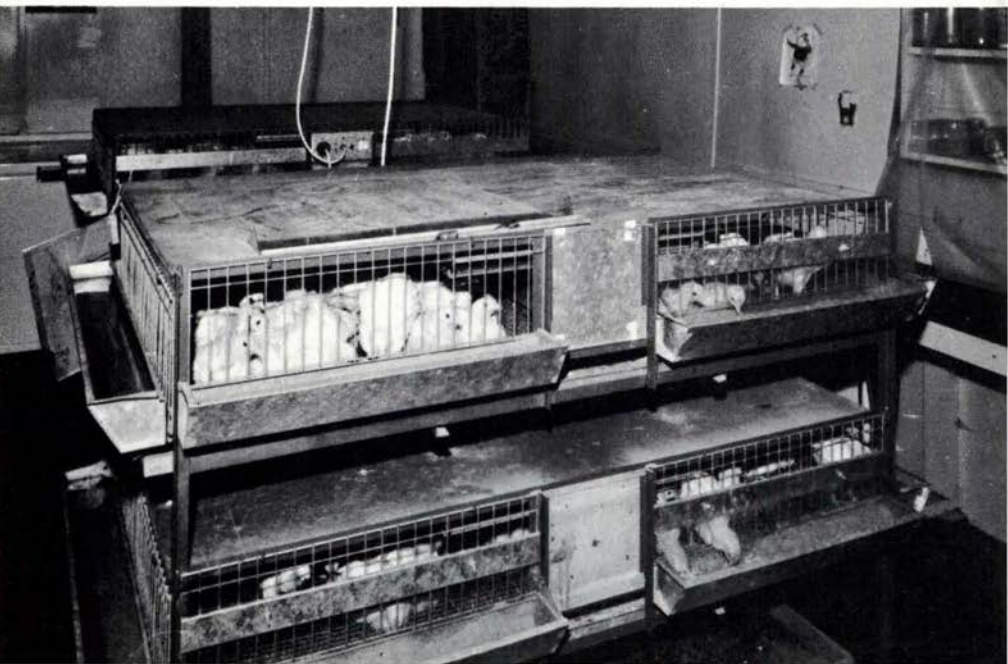
DR R. P. POTTINGER WITH DEGREE II STUDENTS

Miss B. Peddie, a Ph.D. student, weighing chicks dosed with toxic food. This work is supported by the N.Z. Poultry Board.



STUDIES ON MYCOTOXINS IN POULTRY FOOD

Experimental birds housed in incubators in a temperature-controlled room at the Microbiology Department.





A VISIT OF THE NATIONAL RESEARCH ADVISORY COUNCIL

Seated (from left): Dr W. M. Hamilton, Secretary, D.S.I.R.; Dr A. T. Johns, Assistant Director, Department of Agriculture; Prof. B. H. Howard, Professor of Biochemistry; Dr M. M. Burns, Principal; Mr T. D. J. Holderness, Chairman, Lincoln College Council; Mr J. T. Andrews, Chairman, N.R.A.C.; Mr John Boyd-Clark, Lincoln College Council.

Standing: Prof. I. E. Coop, Professor of Animal Husbandry; Dr R. P. Pottinger, Senior Lecturer in Agricultural Zoology; Dr S. N. Adams, Director, Tussock Grasslands Institute; Mr E. D. Holt, N.R.A.C.; Prof. T. W. Walker, Professor of Soil Science; Dr A. P. Mulcock, Reader in Microbiology; Mr A. Henderson, N.R.A.C.; Prof. A. L. Titchener, N.R.A.C.; Prof. A. E. Henderson, Professor of Wool Science; Prof. J. R. Burton, Professor of Agricultural Engineering; Mr H. G. Hunt, Registrar; Prof. R. H. Langer, Professor of Plant Science; Mr W. A. Joiner, Secretary, N.R.A.C.; Mr C. P. McBride, Assistant Secretary, N.R.A.C.; Prof. B. P. Philpott, Professor of Agricultural Economics; Prof. J. D. Stewart, Professor of Farm Management.

MICROBIOLOGY OF FOOD

Micro-organisms play an important role in the manufacture and spoilage of foodstuffs. As well as acting as agents for the spread of disease producing organisms e.g. *Salmonella* sp. An example of this type of research is an investigation into "The microbiology of poultry food", A. P. Mulcock and B. A. Peddie, which has revealed a number.

LEGUME BACTERIOLOGY

Studies on Rhizobium Nitrogen-Fixing Bacteria, I. D. Blair, Mrs V. Price, Miss P. Smythe.

Rhizobia from medics and clovers have been collected from diverse ecological situations where no sheep or cattle are known to have ever grazed. The characteristics of these organisms are being recorded and particular attention is devoted to acid tolerance of strains of *Rhizobium meliloti* and *Rhizobium trifolii*.

Field studies are continuing in tussock grassland situations on the ability of strains of these bacteria to become established under acid soil conditions.

PUBLICATIONS

Blair, I. D., "Studies on Rhizobium Strains, I: *Rhizobium meliloti*". N.Z. J. Agr. Res. (1967) 10: 66-81.

Blair, I. D., "Studies on Rhizobium Strains, II: *Rhizobium trifolii*". N.Z. J. Agr. Res. (1967) 10: 237-52.

Blair, I. D., "Factors Limiting Yields in Wheat". 448, Re-issue Canterbury Chamber of Commerce Bulletin.

Mulcock, A. P., "The Production of Pyocyanine by *Pseudomonas aeruginosa* Isolated from Fleece Wool". N.Z. J. Agr. Res. (1967) 10: 467-70.

Peddie, Barbara (with Keeley G. M. and Hartman, L.), "Determination of Lipo Pexoxides in Meatmeals and Poultry Feeds". N.Z. Sci. 9, 806-8.

AGRICULTURE ZOOLOGY

Studies on Insecticide Resistance, R. A. Harrison, W. P. J. Overmeer

Genetics of resistance in fruit tree mite species to organophosphates, Tedion and kelthane insecticides.

Studies on Insecticide Resistance, II, R. A. Harrison, H. Rudzka

"Fitness" of populations under insecticide pressure using the fruit tree mite (*Tetranychus urticae*) as the laboratory animal.

Studies on Insecticide Resistance, III, R. A. Harrison, S. C. Lee

Development of alternative controls for organo-phosphate resistant fruit tree mites.

Population Dynamics, I: An Ecological Study of the Diamond Back Moth, Plutella maculipennis and the White Butterfly Pieris rapae on Brassica crops in Canterbury, J. W. Ashby.

A study of the population dynamics of these two pests, under field conditions utilizing life tables as the basic tool.

Population Dynamics II: The Population Dynamics of Pasture Insect Pests in Canterbury, R. P. Pottinger, K. V. Ruddenklau

Lifetables are being built for two populations of porina caterpillar, *Wiseana*

cervinata and a study of the adult behaviour and oviposition behaviour of this pest made.

Population Dynamics, III, R. P. Pottinger, J. McLaren

Dynamics of cabbage aphids with special reference to mortality factors.

Flight Behaviour of Argentine stem weevil, Hyperodes bonariensis, R. P. Pottinger, K. V. Ruddenklau

Flight behaviour with particular emphasis to the weather conditions under which flight occurs.

Sampling and Survey Procedures for Lucerne Insects, with particular emphasis on Insects associated with Seed Crops, R. P. Macfarlane

Design of optimal sampling and survey procedures for insects and compilation of a list of the species associated with lucerne crops.

Influence of Earthworms, Allolobophora, caliginosa and grass grubs, Costelytra zealandica, on the availability of pasture soil nitrogen, Othman bin Yaacob
"Open pot" and modified electrolytic respirometer methods were developed to study the influence of these two animals on the availability of soil nitrogen. Mineralization of nitrogen under New Zealand pasture conditions is the result of the interaction of both macro-soil animals and micro-organisms.

Studies in the field of Organophosphorus Chemistry with particular reference to the development of Insecticidal Compounds active against Soil Pests, G. O. Osborne

An investigation of the stability of organophosphorus insecticides in soil is being undertaken in order to determine the nature of the agencies involved in decomposition reactions. This approach is coupled with a programme of synthesis of organophosphorus compounds.

Insect Attractants, Dr C. P. Hoyt, G. O. Osborne

The isolation (from natural sources) identification and uses of chemical substances attractive to insects.

A Study of the Arthropoda associated with Olearia colensoi Hook.f., A. V. Spain

A survey of arthropods in all major *Olearia* stands in N.Z. Special reference to those arthropods damaging *Olearia* stands.

A study of Cryptostigmatid mites in New Zealand, A. V. Spain

A taxonomic study.

Biology of Australian Magpies in New Zealand, J. C. McIlroy

Food habits with special reference to insects of agricultural importance.

Decomposition of Carrion, C. C. Boswell

An ecological study of the decomposition of the carrion of the European Hare *Lepus europaeus* Pallas with special reference to the insect fauna.

Diptera taxonomy, R. A. Harrison

Subantarctic Diptera and New Zealand Muscidae.

PUBLICATIONS

Buchanan, R. S. and Harrison, R. A., "A Laboratory Comparison of the Effects of some Insecticides on the Argentine Stem Weevil (*Hyperodes bonariensis* Kuschel) (Coleoptera: Curculionidae)". N.Z. J. Agr. Res. 10: 82-6, 1967.

Harrison, R. A., "Australian Glow-Worms of the Genus *Arachnocampa* Edwards (Diptera: Mycetophilidae)". Pacific Insects. 8: 877-82, 1966.

- Harrison, R. A., "Insects Pests and Research, I, New Zealand". Span. 10: 114-6, 1967.
- Harrison, R. A., "Entomology in New Zealand". Bull. Ent. Soc. America. 13: 10-12, 1967.
- Pottinger, R. P., "An Ecological Approach to the Study of Insects". N.Z. Agr. Sc. 1:8, 8-15, 1967.
- Wise, K. A. J. and Spain, A. V., "Entomological Investigations in Antarctica" Pacific Insects. 9: 271-93, 1967.

ANIMAL SCIENCE

Nutritional Factors Influencing Fertility in Sheep, I. E. Coop

Large-scale trials are being conducted at Ashley Dene to study the influence of level of nutrition, after conception but during the first two months of pregnancy, upon embryonic survival twinning rate and barrenness as measured at lambing.

Sheep Breeding, I. E. Coop

Interbreeding of the Border-Romney flock is continuing to determine the characteristics and value of this type of sheep after several generations of interbreeding. The ewe flock is now F₃.

Synchronisation of Oestrus, I. E. Coop and V. R. Clark

A series of trials to study the technique and practical usage of ram-induced synchronisation was completed during 1967.

Growth rate of Hoggets during the Winter in the High-Country, V. R. Clark

Comparisons were made of several alternative winter feeds in terms of growth rate through the winter. Anthelmintic and selenium treatments were superimposed upon the feeding treatments.

Influence of Post-Lambing Ewe Nutrition upon Milk Production, V. R. Clark and K. T. Jagusch

Groups of twin bearing ewes were individually fed on high and low levels of nutrition after lambing. Milk production and lamb growth rates were measured.

Gland Transplantations, D. S. Hart

In co-operation with the Medical Research Unit in Christchurch experiments are being conducted in which glands—such as the adrenals, pancreas and ovaries are transplanted into jugulo-carotid loops in sheep. The adrenal transplants are being used for ACTH assay.

Dairy Beef, M. G. Hollard, I. E. Coop

Comparisons in terms of growth rate and carcase quality are being made of the following breeds and crosses—Aberdeen Angus, Hereford, Friesian, Jersey, Jersey Friesian, Red Poll Friesian and Charollais Friesian.

PUBLICATIONS

- Coop, I. E., 1966, "Effect of Flushing on Reproductive Performance of Ewes", J. Agr. Sc. 67: 305-23.
- Coop, J. E., 1966, "The Response of Ewes to Flushing". World Rev. An. Prod., 4: 69-78.
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BIOCHEMISTRY

BIOCHEMISTRY OF RUMEN MICRO-ORGANISMS

(a) "*Quin's Oval*". This unusual microbe, characteristic of the population of the rumen of sheep fed on a sugary diet, has been the subject of further study. Aspects covered include:

(i) *Fermentation of sugars*, Miss B. E. Brough and B. H. Howard.

Lactic acid and volatile fatty acids are the main products of the vigorous fermentation carried out by this organism.

(ii) *Storage polysaccharide*, D. Every and B. H. Howard.

This substance is synthesized in large quantities when the organism is supplied with ample carbohydrate, and is used up in conditions of starvation. It has been shown to be a fairly typical glycogen, such as is synthesized by higher organisms.

(iii) *Ribosomes*, D. C. Reaney (with Dr A. D. Thomson, D.S.I.R., Lincoln).

The properties of these organelles revealed by ultra-centrifugation and electron microscopy are similar to those of bacteria rather than any other group of living things. This confirms the conclusion, based on earlier work in the Department, that "Quin's oval" is an unusual type of bacterium.

(b) *Other rumen micro-organisms*

(i) *Streptococcus bovis*, Miss B. E. Brough and B. H. Howard

The capsular polysaccharide of this bacterium is under investigation. Several oligosaccharides are found in partial acid hydrolysates, and the elucidation of their structures should help to our understanding of the nature of the polysaccharide.

(ii) *Rumen bacteria*, Miss P. Jones and B. H. Howard

A number of strains of highly anaerobic fermentative bacteria are being isolated from the sheep rumen for biochemical study.

PLANT BIOCHEMISTRY

(a) *Asparagine biosynthesis in plants*, M. Lever

Although asparagine was the first amino acid to be identified, and in spite of its importance as the main short-term nitrogen store in many plants, how it is made is still poorly understood. Using lupin seedlings, the biosynthetic pathway is being investigated by enzyme, radioactive tracer and other techniques.

(b) *Cyanide metabolism*, M. Lever

Many soil microbes and pasture plants can synthesize and metabolize cyanide. The pathways involved in its assimilation are being investigated.

VIRUS BIOCHEMISTRY

(a) *Thermophilic bacteriophages*, D. C. Reaney (with Dr A. D. Thomson and M. Reynolds, D.S.I.R. Lincoln)

Varied organic sources have been tested for the presence of viruses capable of reproduction at temperatures above 55°C. The biophysical properties of a new thermophage have been determined. The value of thermophages as a

monitor for the occurrence of thermophilic bacteria in soil has been assessed.
(b) *Genetic circularity and evolution*, D. C. Reanney (with Prof. R. K. Ralph, Department of Cell Biology, University of Auckland)

A hypothesis concerning the relationship between genetic circularity and the evolutionary process has been elaborated.

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FARM MANAGEMENT AND RURAL VALUATION

Development of a computerised cumulative farm accounting and budgetary control system, N. Taylor, N. Gow, R. Tonkin, with B. Clarke, Department of Accountancy

During 1967 a hundred farmers, accountants and stock firm credit managers have co-operated by providing inventory records and monthly transactions. Turn-round of processed material has been disappointing because of programming difficulties. But there is now confidence that an efficient continuous accounting and budgetary control system can be run at a very reasonable cost, if the demand for such a system grows.

Economic and management aspects of intensive pig farming based on meal feeding, O. Kingma

Continuous recording of 14 units is yielding required information on sow performance, feed conversion efficiency, etc., as a basis for economic evaluation of this growing industry in Canterbury.

Economic evaluation of proposed Rakaiā Irrigation Scheme, Mid-Canterbury, N. Taylor, J. Stewart

Little progress has been possible in 1967. Cost/benefit techniques are being applied to individual farm budgets which have been designed to explore the cash flows associated with:

- (a) the maintenance of status quo
- (b) dry land development
- (c) irrigation development

Management aspects of high stocking rates on lucerne on light plains land, J. Stewart

A stocking rate of seven ewes per acre has been achieved on a farmlet basis, and the work is yielding valuable side information on weed control, selenium-copper interaction and wool performance under high lucerne intake.

Evaluation of soil conservation subsidies, G. Frengley

Using one high country station as a case study an economic evaluation of conservation subsidies is being carried out, using cost/benefit techniques.

Economic evaluation of the development of high-country runs, J. Morris

Cost/benefit techniques are being used to study the development of a number of high-country properties.

Credit policies of stock and station agencies, R. Bonifant

A small interview survey has been carried out with the objective of getting information on policy formation and implementation by the agencies.

Development of West Coast Pakihi, R. Bonifant

Development budgets and cost/benefit techniques are being used to evaluate the economics of developing this problem country.

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Lawson, D. B., "A study of the economic and managerial effects of automatic irrigation in the Mid-Canterbury district of New Zealand".

Nuthall, P. L., "A management study of town supply dairy farming in the Christchurch area".

Taylor, N. W., "Management aspects of light plains farming in Canterbury, New Zealand.

HORTICULTURE

Root Distribution in Soil, T. M. Morrison

Using radio-isotopes, the position of roots in the soil profile can be detected. This has been used in following root growth in various vegetable crops.

Studies in Nitrogen-Fixation, T. M. Morrison

(a) Nitrogen -Fixation in Coriariaceae, W. B. Silvester.

The study of the nitrogen fixation process in *Coriaria* using refined techniques for assay of the heavy isotope shows that the first detectable product of fixation after one minute is ammonia.

(b) Nitrogen-Fixation in *Ulex*, T. C. Reid

The fixation process in gorse is being assessed with a view to utilising it in establishment of *Pinus* on low nitrogen areas.

(c) Nitrogen-Fixation in *Alnus*, U. Benecke.

The use of *Alnus* as a cover crop on skeletal soils at high altitude depends on its ability to fix atmospheric nitrogen. The process and amount of fixation under various conditions is being studied.

Mycorrhizas and Nitrogen Nutrition, A. S. Edmonds

The course of nitrogen absorption by mycorrhizas is being compared with that of phosphate and potassium. The utilisation of various nitrogen compounds is also being assessed.

Xylem Flow in Woody Trees, R. W. Heine

The movement of water and dissolved salts is being subjected to a mathematical, physical analysis. There appear to be several pathways of movements of water through the stem. These vary with the type of cell in the stem, its position on the tree, the season of the year and climate.

Phosphate Absorption by Mycorrhizas of Griselinia, W. Thomas

The mycorrhizas of *griselinia* occurs inside the root where the fungus infects many soils. The course of the phosphate movement to the fungus, and its benefit to the host, is being investigated.

Nutrient Transfer in Grafted Tomatoes, S. Subhradrandhu

The movement of root-absorbed nutrients and photosynthetically produced sugars across tomato grafts is being studied. The relative contribution of the stocks and scions is being assessed.

Weedicide Trials in Vegetables, R. A. Crowder

A number of weedicides not yet available commercially are being tested for effectiveness in controlling weeds in a variety of crops.

Direct Seeding of Vegetables, R. A. Crowder

The technique of precision sowing as an aid to extensive production of lettuce, cabbage, onion, celery, tomatoes is being assessed. Results show a marked reduction in time to maturity, compared with transplanting.

Effect of Climate on Vegetable Growth, R. A. Crowder

Detailed wind and temperature measurements are being made and correlated with plant-growth characters.

Seed Size of Carrots and Subsequent Plant Vigour, G. F. Thiele

Seed germination and plant vigour is positively related to seed size.

Tomato Variety Testing and Spacing, G. F. Thiele

Strains selected by D.S.I.R. for resistance to tobacco mosaic virus have been tested for productivity. Spacing trials showed that single plantings gave best returns.

Management in Vegetable and Fruit Production, G. F. Thiele

A study of the factors contributing to economic management of vegetable and fruit holdings has started.

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Morrison, T. M. and Heine, R. W., "Transport of minerals in tree xylem".
Annals of Botany (1966) 30, 807-19.

Morrison, T. M. and English, D. A., "The Significance of mycorrhizal nodules of *Agathis australis*", *New Phytologist* (1967) 66, 245-50.

PLANT SCIENCE

Glutinosa Lucerne, J. G. H. White, G. Meijer

Wide-scale testing of College *Glutinosa* lucerne is in progress in collaboration with the Department of Agriculture and the D.S.I.R. Results to date are encouraging.

Rambler x Glutinosa Lucerne, G. Meijer

A seed multiplication block has been planted. Testing will begin in 1968.

Growth of Lucerne, J. M. Keoghan

The role of the residual leaf area which remains on the plant after cutting is being investigated. Plants with either crown or axillary shoots are compared in their ability to recover from cutting.

Lucerne-Grass Mixtures, R. H. M. Langer, G. Meijer

A range of ryegrass, tall fescue and cocksfoot strains sown with lucerne is under test on light land at Ashley Dene. Two grazing intensities have been imposed on these mixtures in comparison with pure and overdrilled lucerne.

Plant Introduction in Unimproved Grasslands, J. G. H. White, C. G. Janson, W. R. Scott

At Hunua various grasses have been oversown into the resident vegetation. College *Glutinosa* lucerne and subterranean clover strains are being introduced in danthonia and browntop swards in North Canterbury, at Mesopotamia and near Lake Pukaki. These introductions are being studied in relation to aspect, climate and pre-treatment of the vegetation.

Techniques of Lucerne Seed Introduction, J. G. H. White

Pot experiments and field trials are designed to show optimum techniques for the introduction of lucerne in tussock grassland areas. Level of inoculum, lime-pelleting of the seed and use of herbicides on the residual vegetation are the treatments under review.

Cytogenetical Studies of Lucerne, A. G. Fautrier

Phenotypic effects of X-rays, gamma rays and colchicine have been assessed, and selfing of treated plants is in progress. Cytological examination of the effects of these mutagenic agents has been started.

Physiological Basis of Wheat Yield, R. H. M. Langer, A. Ampong, K. Y. Liew, L. B. Olugbemi, M. Hanif

Water stress, extreme temperatures and changes in nitrogen supply imposed at different stages of development are being studied. The morphogenetic effects of these treatments are being followed, especially in relation to the formation of grain in several positions on the spike. Measurements of photosynthesis form part of this programme.

Soya Beans, C. T. Dougherty, O. U. Okereke

Several varieties of soya beans are under test in relation to planting date, row spacing, weed control and irrigation. Special attention is given to the components of yield.

Physiological Ecology of Native Nitrogen Fixers, G. T. Daly

Several native species capable of fixing nitrogen are being studied on a chronosequence of moraines near Mount Cook and on shingle fans in inland

Canterbury. This programme includes field studies and experiments in controlled conditions.

Ecology of Matagouri, G. T. Daly

This investigation is concerned with the responses of matagouri to burning, topdressing, oversowing and heavy grazing. In a pot experiment the competitive ability of matagouri seedlings in grass/clover mixtures is under study.

Ecological Studies, B. P. J. Molloy

(a) Distribution and ecology of prominent weeds such as sweet briar, matagouri, Australian drooping sedge and storksbill.

(b) The relationship between unimproved vegetation and site factors on selected soil sequences in the South Island.

(c) The usefulness of plants as indicators of site conditions, especially soil features and potential land use.

(d) Aspects of pasture ecology in relation to porina and grass grub control.

Nitrogen Nutrition of Pinus, A. S. Edmonds

Gluamine appears to be the main nitrogen component transported in the sap of *Pinus* seedlings. Transport and distribution of nitrogen at different times has been followed by use of ^{15}N .

Seed Production in Tetraploid Ryegrass, L. J. Davies

The effects of time of cutting on subsequent seed production and persistency of 4707 tetraploid Westerwold ryegrass are being studied.

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Daly, G. T., "Nitrogen fixation by nodulated *Alnus Rugosa*". Can. J. Bot. (1966) 44, 1607-21.

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- M. Agr. Sc. Theses:*
- Douglas, J. A., “Establishment and early growth of cocksfoot strains in tussock grasslands”.
- Ampong, A., “Effects of soil moisture stress at different stages of development on the grain yield of two New Zealand wheat varieties”.

SOIL SCIENCE

Analysis of rainfall, A. H. Horn, T. W. Walker

Collection of precipitation along transects across the South Island and analysis for N, S, P, Ca, Mg, K, Na. In co-operation with the Department of Agriculture, trials at each site along one transect from Dunedin to Haast are examining pasture responses to S, P, Mo and Lime, with the aim of correlating atmospheric returns of sulphur and sulphur responses.

Studies on mineralisation of N and S in a climosequence of soils in Central Otago, K. H. Tan

The production of Ammonia, nitrate and sulphate from the microbial breakdown of organic matter in a sequence of seven soils has been studied.

A study of ecosystem development in a chromosequence of P. radiata on sand-dunes in the Manawatu, I. M. Ritchie

The accumulation of C and N in vegetation and soils and changes in nutrient uptake by trees with increasing age.

A study of soils in the Nelson area giving poor growth of P. Radiata, J. D. Kingston, J. A. Adams.

Detailed analysis of soil sequences to elucidate the nature of the problem.

- A study of a sequence of soils in Southland in relation to pasture responses to S. P. K. Mo and Lime*, T. Witchalls
Correlation of soils properties with responses to nutrients.
- A study of a sequence of soils in the Ashburton Gorge area*, G. McFadden.
Correlation of soil properties with responses to nutrients.
- Ecosystem Development*, A. S. Campbell, E. J. B. Cutler
A study of the changes in mineralogy during soil development.
- Ecosystem Development*, R. T. Baker
A study of the changes in the nature of organic phosphorus compounds during soil development.
- Postassium status of Canterbury soils*, A. F. R. Adams
Analysis of herbage and soils from field experiments carried out by the Department of Agriculture (R. Stephen).

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- Heine, R. W., "The semester—a Lincoln experiment." Annual Review, Lincoln College, 1966.
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VETERINARY SCIENCE

Plant-Animal Relationships on Pure-species Pastures, J. W. McLean, G. G. Thomson and B. M. Lawson.

- (a) Studies on the flavour and odour of the cooked fat and lean meat of lambs raised on different pasture species.

This work was carried out in association with the Animal Chemistry Division of D.S.I.R. in Wellington, where the attributes of flavour and odour in casserole dishes were measured by a trained taste panel. The panel were able regularly to identify the somewhat more pronounced "mutton flavour" of the meat and fat of lambs grazed for about eight weeks on lucerne as contrasted with that of lambs grazed on perennial ryegrass. Untrained personnel in average households of the College Staff were unable to distinguish between the two classes of meat when cooked and served as hot roast lamb.

- (b) Phyto-oestrogen assays

Data from trials on phyto-oestrogen activity of grasses and legumes are being analysed. The increase in teat length in young ewe and wether lambs showed good conformity with the lamb uterine weight assay technique.

(c) Carcass weight and composition studies (with A. M. Nicol). The Prediction of Carcass Composition of lambs from live weight and red cell volume.

From these studies with this particular class of lambs, it was concluded that, although it was possible statistically to improve the prediction of fat-free carcass weight from the live weight by measuring either red cell volume or live weight gain, the improvement was of little practical significance. This, in effect, was due to the very high correlation between live weight and fat-free carcass weight.

(d) Work on the effects of different pasture species on wool growth, the histology of the thyroid, adrenals and pancreas, liver and kidney size and the rumen microbial population is continuing.

Control and Eradication of Brucella Ovis Infection in College Stud, G. G. Thomson and B. M. Lawson

For the third consecutive year the testicles of the ram hoggets were palpated and those with abnormal testicles were removed from the flock. A random sample of 50 rams were blood tested by Mr Ris of Wallaceville using the complement fixation and haemagglutination tests. All were negative. Since no organisms have ever been isolated and the blood tests were all negative, we assume the flock to be free of *Brucella ovis* infection.

Control of Footrot in Sheep by Frequent Administration of Copper, G. G. Thomson and B. M. Lawson.

To assess the efficacy of this method, a trial mob of 750 Corriedale ewes was made available and copper glycinate was administered to one group of 175 ewes three times in five months. The feet of all ewes were examined every six weeks. No footrot occurred in any of the groups, thus making evaluation impossible.

The Effect of using Selenium and Copper on Ewes at Ashley Dene and on Calves at the College Dairy Farm, G. G. Thomson and B. M. Lawson

Four groups, each of 175 Corriedale ewes, were given either copper, selenium, copper and selenium (or nothing) at frequent intervals throughout the winter and the effects recorded. The administration of selenium and copper reduced neonatal lamb deaths, prevented Enzootic Ataxia and White Muscle disease and increased the growth rate of lambs from the treated ewes. The administration of selenium to ewes grazing on a copper deficient pasture appeared to increase the copper level in the blood of such ewes and also to increase the liver copper of their lambs.

Selenium and copper given to calves on the College Dairy farm gave increased weight gains in the bucket-fed calves.

Thyroid Function Studies in Cattle and Sheep, C. H. C. Irvine and B. M. Lawson.

Thyroid function in cattle and sheep was assessed from thyroxine secretion rate measured by the isotope dilution technique. Thyroid function in the pregnant and lactating ewe is much higher than in any other animal so far studied. Iodine metabolism was investigated using radioactive iodide. Among the unexpected and interesting findings were (a) the very low level of urinary iodine in sheep, often 3% of a dose of radioactive iodine being excreted in urine compared with 40-80% in other species; (b) the very high level of iodine in milk, 30-80 times that of plasma iodide compared with 2-10 times in other species. The nature and significance of milk iodine, and the nature and binding of plasma iodine compounds, are being studied. A method for measuring plasma thyroxine in these species is being developed, as a prerequisite to free thyroxine determination.

Brucellosis Control, C. S. M. Hopkirk and C. Robertson

Composite milk samples are taken from all herds in the Christchurch supply area and subjected to the "ring" test. In any herd giving a positive reaction, milk samples are taken from the individual cows and tested by the "ring" and whey agglutination tests. Positive reactors are slaughtered. The Christchurch milk supply area is thus kept free of brucellosis.

Mastitis Control, C. S. M. Hopkirk and C. Robertson

In 1966, an experimental herd of heifers was established to study mastitis control measures that could be applied easily by the dairy farmer. The procedure is thoroughly to wash the udder with running water, to jet the cups with water after milking each cow, to apply antiseptic cream to the teat surface and orifice and to avoid overmilking. By these methods mastitis is being controlled as judged by clinical examination and by leucocyte assessment and laboratory culture twice a week. In addition, the effect of mastitis on the non-fatty solids of the milk is being studied.

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Nicol, A. M., Thomson, G. G. and McLean, J. W. (1968), "The Prediction of Carcass Composition of Lambs from Live Weight and Red Cell Volume". N.Z. J. Agr. Res. (in press).

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WOOL SCIENCE

Application of Progeny Test and Half-classing Methods in Merino Ram Selection, A. E. Henderson

The purpose of this work is to test the application of sheep selection methods suggested by Australian research workers. Each year wool production of year-old Merino rams is measured and the sheep ranked according to score. The top 20% are retained for a further year and production again measured. Year and two-year-old wool scores are positively correlated. Final stud selections are made at two years.

Variability of Crossbred Wools

It is common opinion that interbreeding of crossbred sheep leads to increase in variability of the fleece. Measurements are being made on wool from two generations of such interbred sheep.

Fibre Diameter/Crimp Frequency Relationships in Coarse Wools

Crimp is an important characteristic of carpet wools. A study is in progress

of the nature of and variability of the fibre diameter/crimp frequency relationship.

Fleece Development of Border Leicester Sheep, A. E. Henderson

This study is designed to give information on the timing of important stages in fleece development.

Production and Analysis of Yellow Wool, A. E. Henderson

A special flock of sheep noted for production of discoloured wool has been established. This flock is to serve as a source of experimental material for a number of College Departments and the Wool Research Organisation. Studies are being made of fleece architecture, microbiology and chemistry associated with discoloration.

PUBLICATIONS

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